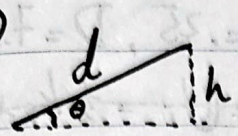
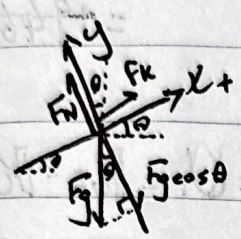
HW #11

1)



Given: $F_g = 267N$, $d = 6.10m$, $\theta = 20°$, $k = .10$, $V_i = .457 m/s$, $\Delta E_{mech} = 0$



a) $\Delta E_{ther} = |F_k \cdot d|$

$= |F_N \cdot k \cdot d|$

$= |F_g \cos\theta \cdot kd|$

$= |-267N \cdot \cos 20° \cdot .1 \cdot 6.1|$

$= 153.05 J$

b)

$h = d\sin\theta$

$m = F_g / g$

$\Delta E_{mech} = \Delta K + \Delta U + \Delta E_{th}$

$-\Delta E_{th} = K_f - K_i + U_f - U_i$

$= \frac{1}{2}mV_f^2 - \frac{1}{2}mV_i^2 + 0 - mgh$

$-\frac{2\Delta E_{th}}{m} = V_f^2 - V_i^2 - 2gh$

$V_f^2 = -2\frac{\Delta E_{th}}{m} + V_i^2 + 2gh$

$= -2\frac{\Delta E_{th} g}{F_g} + V_i^2 + 2gd\sin\theta$

$= -11.25 + .21 + 40.93$

$V_f = \sqrt{29.89}$

$= 5.47 m/s$

2) Given: $m = 3.50\,kg$, $k_s = 640\,N/m$, $\mu_k = .25$, $D = 7.80\,m$

a) $\Delta E_{th} = F_k \cdot d$

$= \cancel{mg \cdot d}$

$= mg \cdot d \cdot k$

$= 267.8 \cdot .25$

$= 66.95\,J$

b) ~~$K_{max} = U_i = \frac{1}{2} k_s x^2$~~

~~$= 19468.8\,J$~~ X

b) $K_{max} = U_i = K_f - K_i + U_f + E_{th}$

$= 0 - 0 + 0 + E_{th}$

$= 66.95\,J$

c) $U_i = \frac{1}{2} k_s x^2$

$x^2 = \frac{2U_i}{k_s}$

$x = \sqrt{\frac{2U_i}{k_s}}$

$= .457\,m$

3)

Given: $m = 25\,kg$, it is a baby bear.

$d = -12$, $V_f = -5.6\,m/s$

a) the Potential Energy transfer into the Kinetic Energy & Thermal Energy.

b) $K_f = \frac{1}{2} m V_f^2$

$= \frac{1}{2} \cdot 25\,kg \cdot (-5.6\,m/s)^2$

$= 392\,J$

c) $\Delta E_{mech} = K_f - K_i + U_f - U_i + \Delta E_{th}$

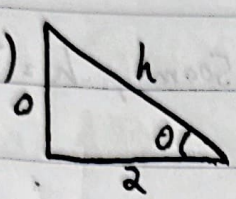
$0 = 392\,J - 0 + 0 - mgd + F_k d$

$F_k d = mgd - 392\,J$

$F_k = (mgd - 392\,J)/d$

$= 2551\,J / 12.0\,m$

$= 212.58\,N$

4) 

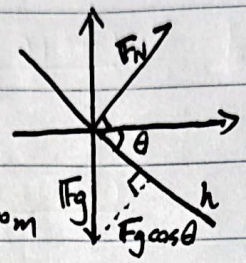
Given: $m = 520\,kg$, $o = 300\,m$, $h = 500\,m$, $\mu_k = .25$, $\sin\theta = .6$

b) $E_{th} = f_k \cdot d$

$= \mu_k \cdot F_N \cdot h$

$= .25 \cdot 4080.96\,N \cdot 500\,m$

$= 510120\,J$



$F_N = F_g \cos\theta$

$= mg \cdot \sqrt{1-.6^2}$

$= 520\,kg \cdot 9.81\,m/s^2 \cdot .8$

$= 4080.96\,N$

a) $\Delta E_{mech} = U_{gf} - U_{gi} + K_f - K_i + E_{th}$

$0 = 0 - U_{gi} + K_f - \cancel{K_i} + E_{th}$

$U_{gi} = K_f + E_{th}$ $\longrightarrow$ c) $K_f = U_{gi} - E_{th}$

$= mgy$ ~~$+E_{th}$~~

$= 520\,kg \cdot 9.81\,m/s^2 \cdot 300\,m$

$= 1530360\,J$

$= 1530360 - 510120$

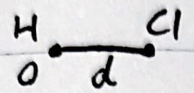
$= 1020240\,J$

d) $K_f = \frac{1}{2} m v^2$

$v = \sqrt{2K_f/m}$

$= 62.64\,m/s$

5) a)



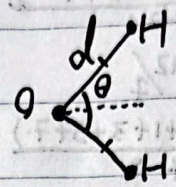
$$m_{cen} = \frac{m_H \cdot d_H + m_{Cl} \cdot d_{Cl}}{m_H + m_{Cl}}$$
$$= 1.264 \times 10^{-10}\ m$$

Given: $d = 1.30 \times 10^{-10}\ m$,

$m_H = 1u$,

$m_{Cl} = 35u$

b)



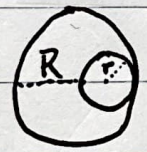
Given: $\theta = 105°$, $d = 1 \times 10^{-10}\ m$, $m_O = 16u$, $m_H = 1u$

$$m_{cen}\hat{j} = 0$$

$$m_{cen}\hat{i} = \frac{m_O \cdot d_O\hat{i} + 2 \cdot m_H \cdot d_H\hat{i}}{m_O + 2 \cdot m_H}$$
$$= \frac{2 \cdot 1u \cdot \cos(\theta/2)}{18u}$$
$$= 6.81 \times 10^{-12}\ m$$

$$m_{cen} = 6.81 \times 10^{-12}\ m\,\hat{i} + 0\,\hat{j}$$

6)



Given: $r = R/2$, $\vec{r}_{cenR} = 0$, $\vec{r}_{cenr} = R/2$

$$m = V\rho$$
$$= l \cdot w \cdot h \cdot \rho$$
$$m = l \cdot w \cdot \sigma$$
$$= A\sigma$$

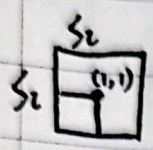
$$\cancel{\vec{r}_{cen}\hat{i} = \frac{mR(0) + -mr(\frac{R}{2})}{mR - mr}}$$
$$\cancel{= \frac{R(0) - r(\frac{R}{2})}{R - r}}$$
$$\cancel{= \frac{0\pi R^2\sigma - \frac{R}{2}\pi\frac{R^2}{4}\sigma}{R - \frac{R}{4}}}$$
$$\cancel{= \frac{}{3R/4}}$$

$$\cancel{mR(0) = \pi R^2\sigma(0)}$$
$$\cancel{mr(\tfrac{R}{2}) = \pi\tfrac{R^2}{4}\sigma(\tfrac{R}{2})}$$
$$m_R = \pi R^2\sigma$$
$$m_r = \pi\frac{R^2}{4}\sigma$$
$$= \frac{m_R}{4}$$

$$\vec{r}_{cen}\hat{i} = \frac{mR(0) - m_r(\frac{R}{2})}{mR - mr}$$
$$= \left(0 - \frac{m \cdot R}{8}\right)\left(\frac{4}{3m}\right)$$
$$= -\frac{R}{3}$$

$$\vec{r}_{cen} = (-R/3\,\hat{i},\ 0\,\hat{j})$$

7) Given: $S_L = 2\,cm$, $S_m = 1\,g$



a) $\vec{r}_{cen}\hat{i} = \frac{2\cdot(1+1+1+1+3+5)}{7\cdot 2}$

$= \frac{24}{14} = 12/7$

$\vec{r}_{cen}\hat{j} = \frac{2\cdot(1+1+1+3+5+7)}{7\cdot 2}$

$= 18/7$

$\vec{r}_{cen} = (\frac{12}{7}\hat{i}, \frac{18}{7}\hat{j})$

b) $\vec{r}_{cen}\hat{i} = \frac{2\cdot(-1-1-3+1+1+1+3)}{2\cdot 8}$

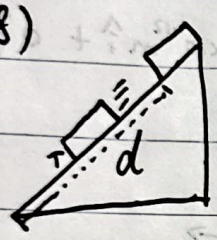
$= 1/4$

$\vec{r}_{cen}\hat{j} = \frac{2\cdot(1+1+1+1+3+5+7+7)}{2\cdot 8}$

$= 26/8 = 13/4$

$\vec{r}_{cen} = (\frac{1}{4}\hat{i}, \frac{13}{4}\hat{j})$

8)

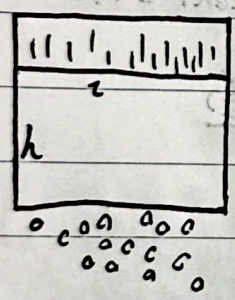


$m = VP$

$= lhT\rho$

$V = d/t$

$= \cancel{1/5.5}$



$P = mV$

$= lhT\rho\, d/t$

$= \cancel{5km}\ 500m \cdot 100m \cdot .4m \cdot 350kg/m^3 \cdot 1Km/5.5s$

$= 20000\,m^3 \cdot 350kg/m^3 \cdot 1km/5.5s$

$= 7000\,kg\,km/5.5s$

$= 1272.727\,kg\,km/s$

Given:

$T = .4m$,

$d = 1km$,

$l = 500m$,

$h = 100m$,

$t = 5.5s$,

$\rho = 350kg/m^3$